

Idea Propuesta Trabajo de Grado

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22 de agosto de 2025



UNIVERSIDAD

- 1 Objetivo de estudio
- 2 Metodología utilizada

- **Objetivo General:** Desarrollar un modelo de predicción de suscripción de Convenios de Doble Tributación y su impacto fiscal en el recaudo del impuesto sobre la renta en Colombia
- **Artículos base:**
 - [1] Who will sign a double tax treaty next? A prediction based on economic determinants and machine learning algorithms (2024) *Economic Modelling*
 - [2] Exploring the Impact of Double Taxation Treaties on India's FDI: A Panel Data Approach Using the Gravity Model (PrePrint) (2025) *Research Square*
 - [4] A gravity model analysis of international migration from BRIC to OECD countries using Poisson Pseudo-maximum likelihood Approach (2021) *Heliyon*

- Predicción de enlaces temporales en redes complejas [5] *Temporal Link Prediction: A Unified Framework, Taxonomy, and Review. Qin and Yeung (2023)* [6] *International trade evolution and competition prediction of boron ore: Based on complex network and link prediction. Zhu, Zhou, Zhang, and Sun (2023)*
- Modelo gravitacional [1] [4]
- Pronostico de variable de recaudo para el impuesto sobre las ventas. [3] *Forecasting Tax Revenue in Indonesia using Single and Hybrid Methods. Jayanti and Selo and Paulus (2025)*

- Bases de datos del DIAN
 - Comercio Internacional
 - Bases de datos de precios de transferencia de contribuyentes de países con y sin convenios (a solicitar)
- Bases de datos DANE y otras administraciones
 - PIB
 - Población
- Bases de datos construidas
 - Bases de convenios internacionales de los países con los que Colombia tiene acuerdo
 - bases de datos con elementos geográficos, demográficos, culturales y de comercio de otros países



Dmitry Erokhin and Martin Zagler.

Who will sign a double tax treaty next? a prediction based on economic determinants and machine learning algorithms.

Economic Modelling, 139, 2024.



Firdous Ahmad Hurrah and Khalid Ashraf Chisti.

Exploring the impact of double taxation treaties on india's fdi: A panel data approach using the gravity model.

Research Square, 2025.



Desy Jayanti, Selo Sulisty, and Paulus Insap Santosa.

Forecasting tax revenue in indonesia using single and hybrid methods.

In *2025 International Conference on Advancement in Data Science, E-learning and Information System (ICADEIS)*, pages 1–6, 2025.



Wajiha Manzoor, Sajid Hussain, and Muhammad Zeeshan Khan.
A gravity model analysis of international migration from bric to oecd countries using poisson pseudo-maximum likelihood approach.
Heliyon, 7:e07357, 2021.



Meng Qin and Dit-Yan Yeung.
Temporal link prediction: A unified framework, taxonomy, and review.
ACM Computing Surveys, 56:1–40, November 2023.



Mingxue Zhu, Xuanru Zhou, Hua Zhang, Lu Wang, and Haoyu Sun.
International trade evolution and competition prediction of boron ore:
Based on complex network and link prediction.
Resources Policy, 82:103542, 2023.